The Long Shadow of Mission Medicine

Colonial missionary healthcare, institutional trust and the spatial burden of disease in Sub-Saharan Africa

Michael Chanda Chiseni
[Affiliation]

Research seminar | [Venue, date]

The puzzle: care within reach, but often unused

What we observe

- Facilities are often within physical reach, yet antenatal care, facility delivery and vaccination remain low (Peters et al. 2008)
- Disease burden is highly uneven within countries, not only between them (Weiss et al. 2019; LBD Collaborators 2020)
- Communities similar in income, education and distance still differ in how readily they use formal care (Ensor and Cooper 2004)

[Insert map: within-country variation in facility births or DTP3 by DHS cluster, with health facilities overlaid]

Why do otherwise similar communities engage so differently with formal health systems?

Health-systems research: trust matters, but is treated as contemporary

What the literature shows

- Trust is constitutive of health care as a social institution: it shapes whether people present early, disclose, follow advice and return (Gilson 2003, 2005)
- Where trust is low, people delay care, bypass facilities, default on treatment and turn to informal providers (Østergaard 2015; Arakelyan et al. 2021)
- Faith-based providers remain major actors in African health systems (Olivier et al. 2015)

What it leaves out

- Trust is explained by present conditions: quality, cost, respect, recent experience
- No historical account of why trust varies so much across space
- Communities' earlier encounters with formal medicine are left out

This project: a historical explanation for spatial variation in trust in health systems

Economic history: colonial legacies persist, but the health channel is untested

What the literature shows

- The slave trade left lower trust today (Nunn and Wantchekon 2011)
- Coercive colonial medical campaigns lowered trust in medicine and vaccination (Lowes and Montero 2021)
- Proximity to colonial medical missions in India predicts better health today (Calvi and Mantovanelli 2018)
- Mission health facilities lower HIV today; mission settlements in general raise it (Cagé and Rueda 2020)
- Missions are linked to lower interpersonal trust in former indirect-rule colonies (Okoye 2021)

What remains open

- Studies measure outcomes, not the pathway of engagement with health systems
- Mission medicine has not been mapped at scale across British Africa
- Most rely on mission atlases, which omit 76 to 91% of missions (Jedwab, Meier zu Selhausen and Moradi 2022)

No study links mission medicine to trust in health systems today, and through it to disease burden

Historical argument (1): missions were the first formal health system

Hospitals in Northern Rhodesia

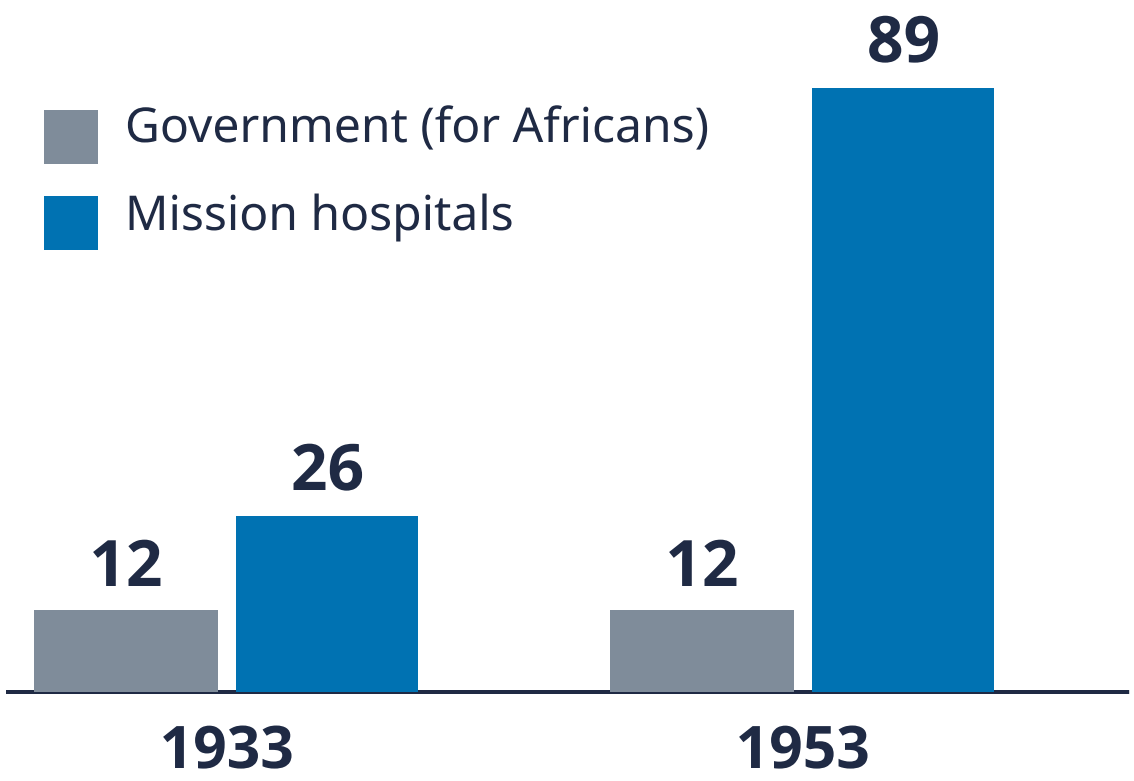


Chart: Chiseni (2025), African Studies 84(1–2), Figure 3 and text

The first formal health system

- Missionary societies supplied hospitals, dispensaries and trained staff decades before colonial states built public health systems (Vaughan 1991; Hokkanen 2019; Doyle et al. 2020)
- In Northern Rhodesia the government ran 12 hospitals for Africans; mission hospitals grew from 26 in 1933 to 89 in 1953, mostly rural (Chiseni 2025)
- African auxiliaries carried mission medicine forward and reinterpreted it locally (Kalusa 2007)

Historical argument (2): the encounter was uneven

Trust dividend

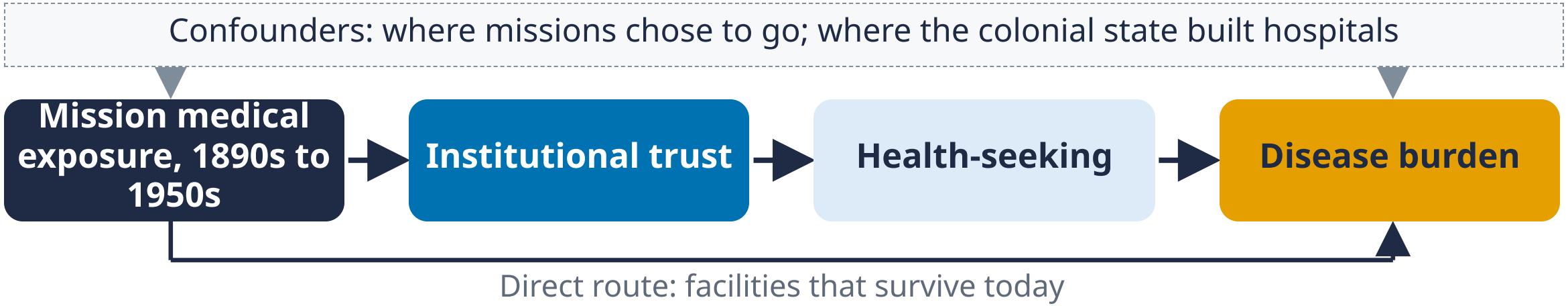
- At Kalene Hospital, patients negotiated how they used mission medicine (Kalusa 2014)
- In Tanganyika, mission medicine built familiarity with formal treatment (Jennings 2008)
- In colonial India, medical missions left lasting gains in health (Calvi and Mantovanelli 2018)

Trust deficit

- Tsetse and sleeping-sickness control in Northern Rhodesia fed rumours of blood-taking (White 1995)
- Missions made church membership conditional on renouncing female circumcision; Kikuyu Christians founded independent churches and schools (Anderson 2018)
- Leprosy settlements isolated patients in the service of a 'civilised', Christian Uganda (Vongsathorn 2012)

Argument: the geography of mission medicine laid down a lasting substrate of institutional trust, positive in some places and negative in others

Four research questions, one causal chain



RQ1 Map
Where was mission health care, facility by facility, by society, territory and period?

RQ2 Link
Do exposed communities differ in trust and care-seeking today, and does that predict burden?

RQ3 Channels
Surviving facilities, health-worker norms, religious norms, or memories of coercion?

RQ4 Meaning
How do communities understand their own relationship with formal care?

Data (1): a facility-level map of mission medicine

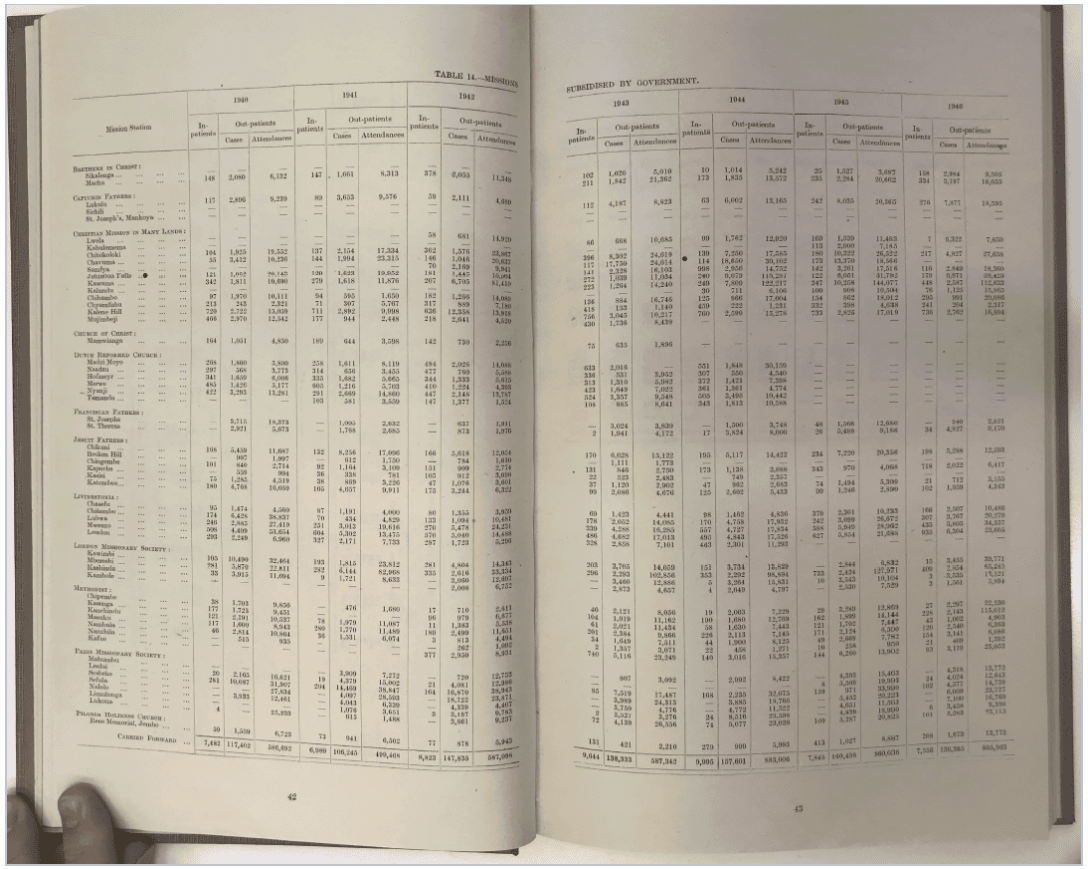


TABLE 14.—MISSIONS SUBSIDISED BY GOVERNMENT.

Northern Rhodesia medical report, Table 14: subsidised mission stations, in- and out-patients, 1940 to 1946

76 to 91%

of missions are missing from commonly used atlases (Jedwab, Meier zu Selhausen and Moradi 2022)

Blue Book ecclesiastical returns

Every mission station, by society and year

Colonial Medical Reports

Mission and government hospitals and dispensaries, with annual in- and out-patients

Validation

Against existing mission data and my Zambian benchmark (Cagé and Rueda 2016; Hedde-von Westernhagen and Becker 2022; Brown 2025; Chiseni 2025)

Data (2): linked to contemporary surveys, facilities and burden maps

Source	What it measures	Level
IPUMS-DHS (Minnesota Population Center 2022)	Antenatal care, facility births, vaccination, care-seeking; refusal of free blood tests; child mortality	Individual; displaced cluster
Afrobarometer Round 9 (2021–23), Round 10 (2024)	Trust in Ministry of Health vs police, courts (R10); respect at clinics, COVID-19 vaccine trust (R9)	Individual; region
IHME Local Burden of Disease	Modelled malaria and child-health burden, 5 × 5 km	Grid cell; district
KEMRI-Wellcome / WHO facilities (Maina et al. 2019)	98,745 public-sector facilities; faith-based ownership where recorded	Facility
Colonial state hospitals (Bolt and Cilliers 2025)	Government hospitals in 12 British colonies	Facility; territory
Fieldwork, four Zambian communities	About 50 interviews and 8 single-sex focus groups	Community

Methodology (1): separating mission medicine from mission placement

The problem

Missions went first to healthier, more accessible, richer places.

Atlas-based mission effects shrink sharply with fuller data and controls.

Jedwab, Meier zu Selhausen and Moradi 2022

1. **Control for placement and state provision**
   Malaria, railways, cash crops, urban access (Jedwab et al. 2022); colonial state hospitals (Bolt and Cilliers 2025); country fixed effects
2. **Medical versus non-medical stations**
   Same selection into location, different medical exposure (Calvi and Mantovanelli 2018)
3. **Societies' medical propensity as an instrument**
   Leave-one-out share of a society's stations with medical facilities (Calvi and Mantovanelli 2018; Cagé and Rueda 2016)
4. **Dose-response, distance decay and robustness**
   Oster (2019) bounds; plausibly exogenous IV (Conley, Hansen and Rossi 2012); spatial noise (Kelly 2019)

A bounded estimate, not a claim of perfect exogeneity

Methodology (2): measuring trust and decomposing the effect

Measuring trust

- Revealed: refusal of free DHS blood tests (Lowes and Montero 2021)
- Stated: trust in the Ministry of Health against police and courts (Afrobarometer Round 10)
- Experienced: respect, bribes and absent staff at clinics (Afrobarometer Rounds 9 and 10)
- Lived: fieldwork accounts of using, bypassing or avoiding clinics

Decomposing the effect

- The total effect of mission medical exposure is the primary estimand
- Controlled direct effect, holding trust and behaviour fixed, by sequential g-estimation (Acharya, Blackwell and Sen 2016)
- Sensitivity to confounding between mediator and outcome (Imai, Keele and Tingley 2010)
- Individual-level links within the DHS; across surveys at a common geography

Today's facilities and care quality are channels, not controls

Methodology (3): telling the channels apart

Channel	What you'd see	Main evidence
Surviving faith-based facilities	Effect concentrated where the mission facility still runs	Facility database; DHS place of delivery
Health-worker norms	Survives closure; clinic care follows the mission's history	Training records; DHS and Afrobarometer care quality
Religious norm	Strongest among members of the mission's church	Religion in DHS and Afrobarometer; church teaching
Coercion	Lower trust in medicine even where care is no worse	Archival coding; health versus police trust; memories

Care quality: better after mainly care, possibly harsher after coercion. Coercion test after Lowes and Montero (2021).

If the answer is yes: part of Africa's health inequality is historical

For explanation

A historically rooted, demand-side share of within-country health inequality that supply-side explanations miss (cf. Peters et al. 2008; Gilson 2003)

For policy

Where low use reflects a trust deficit, not distance or cost, infrastructure alone will not close the gap. Locally recruited health workers, partnerships with faith-based providers (Olivier et al. 2015) and respectful care matter more.

If the answer is no

The gap is mainly a present-day problem, which is also worth knowing. Either way, the first open, facility-level map of mission medicine in British Africa.

If yes: trust is a lever that infrastructure spending alone will not move

Where I'd welcome your feedback

Coverage

Which territories and years of the medical reports are complete enough for a core panel?

Identification

Is societies' medical propensity a credible instrument, given doctrinal differences?

Trust

Which measure best captures trust in health systems, and how should they be combined?

Fieldwork

How should the four Zambian sites be chosen to separate the channels?

Michael Chanda Chiseni | [email]

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